“RESPONSIVE QUIZ WEBSITE”

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A Project Report Submitted to

Government Girls Polytechnic, Lucknow

Partial fulfilment of the requirement for the award of

The degree of

Diploma in Information technology

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ACKNOWLEDGEMENT

It is with immense pleasure that I submit this project report on The **Responsive Quiz Website**, which was successfully developed during my **Summer Internship**.

I am deeply indebted to **our Head of Department, Miss Priyanka Mam**, for her mentorship, critical suggestions, and invaluable insights. Her unwavering support and guidance were instrumental in the successful completion of this work.

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CERTIFICATE

This is to certify, that ***Kiran Verma (E24208035600004)***

has successfully completed her project work entitled

RESPONSIVE QUIZ WEBSITE

as per the syllabus and have submitted a satisfactory report

on this project as a part offulfilment towards degree of

Diploma in Information Technology

from

Government Girls Polytechnic, Lucknow.

External Sign **HOD Signature**

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ABSTRACT

This project details the development of a **Responsive Quiz Website** designed to provide an accessible, interactive, and engaging assessment tool for online learning. The primary objective was to overcome the limitations of static assessment methods and non-mobile-friendly interfaces prevalent in many existing educational platforms.

The application was built entirely using **front-end technologies: HTML5, Cascading Style Sheets (CSS3), and JavaScript (JS),** ensuring lightweight operation and high compatibility. The core functionality includes dynamic question loading, instant scoring, and end-of-quiz feedback, all delivered through a robust, responsive design.

The website successfully adapts its layout and functionality seamlessly across diverse devices, including desktops, tablets, and smartphones, thereby maximizing user reach and convenience.

In conclusion, the Responsive Quiz Website offers a practical, scalable, and user-centric solution for conducting effective online self-assessment and training.

INTRODUCTION

A ***Quiz Website*** is an ***interactive web-based*** application designed to test user knowledge through multiple-choice questions. The purpose of this project is to develop a modern, responsive, and user-friendly quiz system where users can select question categories, answer timed questions, and view their scores at the end.

The quiz automatically highlights correct and wrong answers, supports a countdown timer, and features an attractive ***UI built using HTML, CSS, and JavaScript.***

OBJECTIVES OF THE PROJECT

To implement a category-based quiz
✔ To create a countdown timer for each question
✔ To highlight correct and incorrect answers
✔ To store quiz logic in JavaScript
✔ To enhance user experience with modern UI using CSS

SCOPE OF THE PROJECT

The quiz system can be used:

✔ In educational institutions for self-assessment
✔ For practice tests in competitive exams
✔ As part of online learning tools

System Requirement

Software Requirements

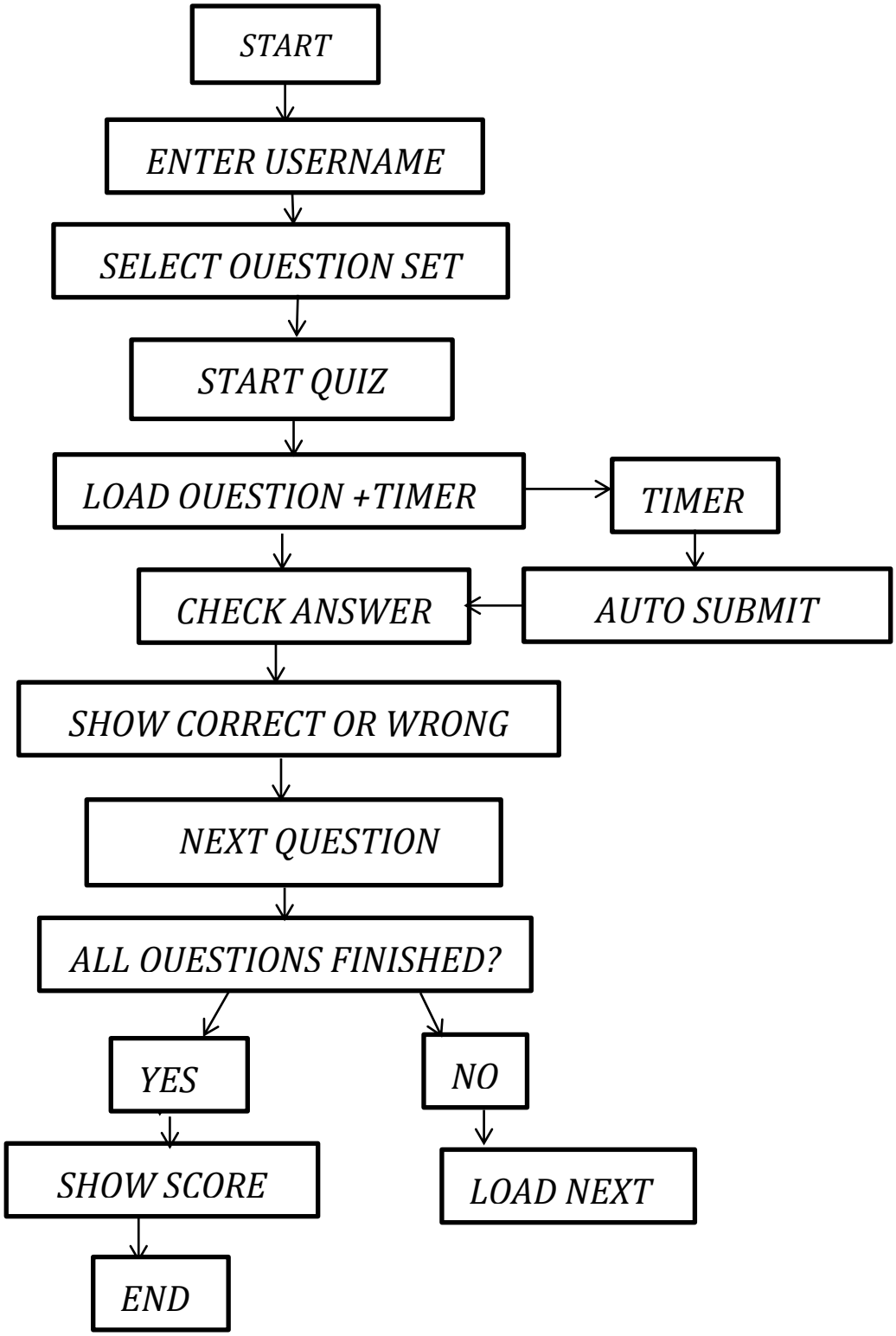
- ❖Windows/Linux operating system
- ❖Any browser (Chrome / Edge / Firefox)
- ❖VS Code text editor
- ❖HTML5, CSS3, JavaScript ES6

Hardware Requirements

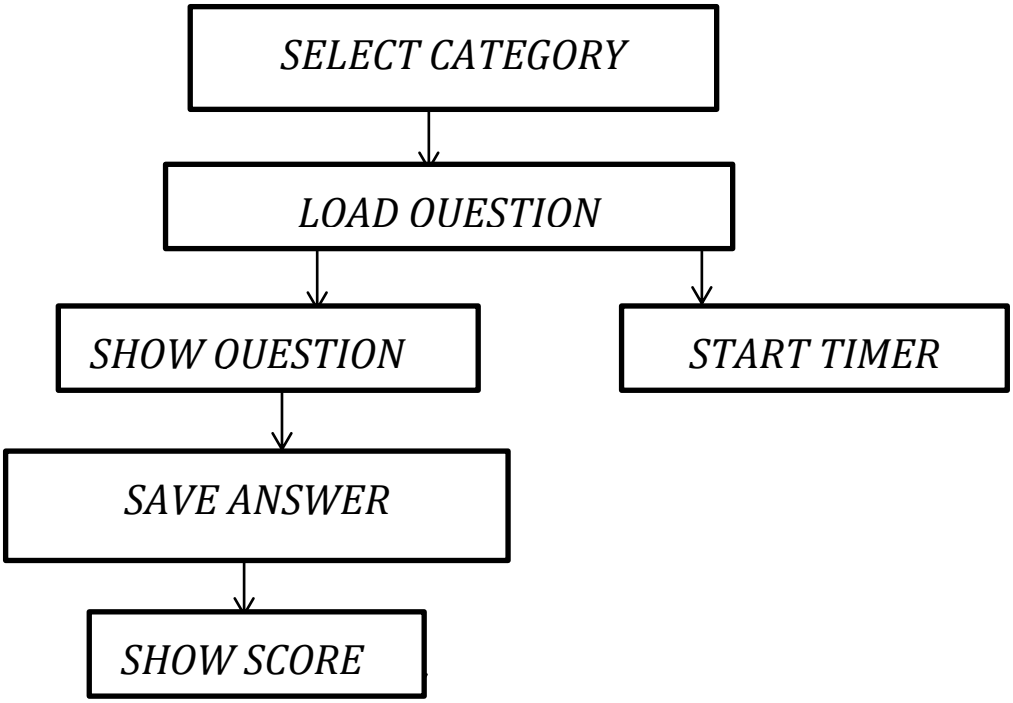
- ❖Minimum 4GB RAM
- ❖1.6 GHz processor
- ❖Keyboard + Mouse

SYSTEM ANALYSIS AND DESIGN

FLOWCHART



DATA FLOW DIAGRAM



MODULES OF THE PROJECT

Module 1: User Input Module

- Enter username
- Select question set
-

Module 2: Question Display Module

- Loads question and options dynamically
- Displays current question index

Module 3: Timer Module

- 15-second countdown per question
- Auto-submit when timer ends

Module 4: Answer Checking Module

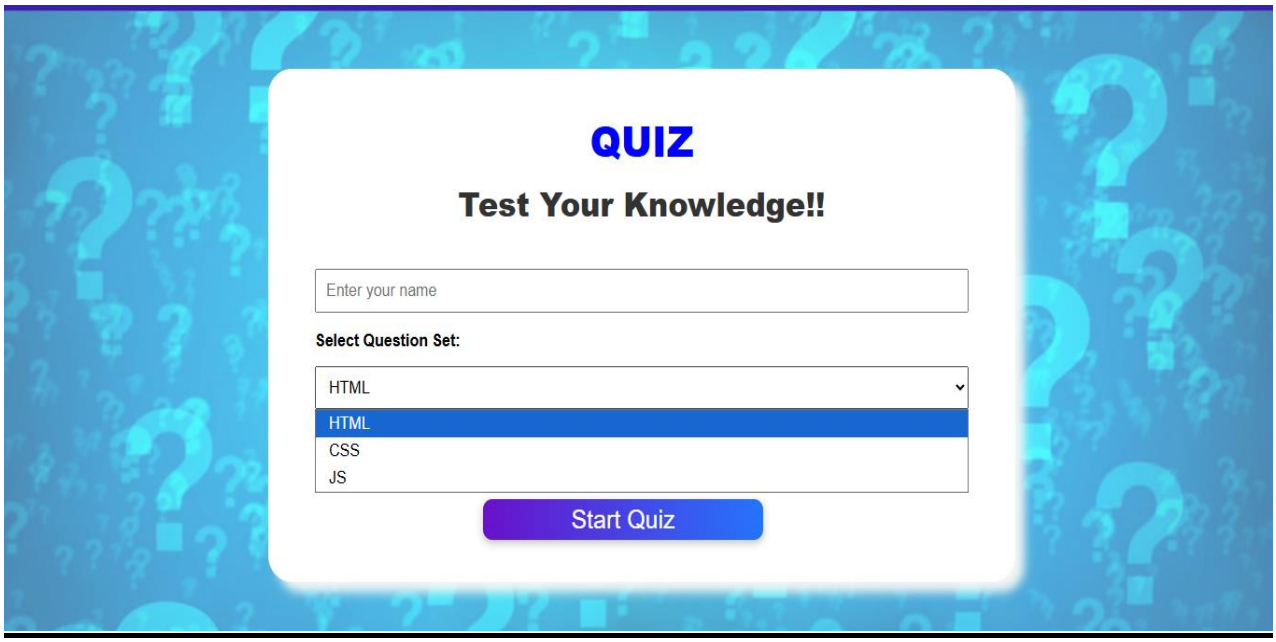
- Highlights correct/wrong answers
- Maintains score

Module 5: Result Module

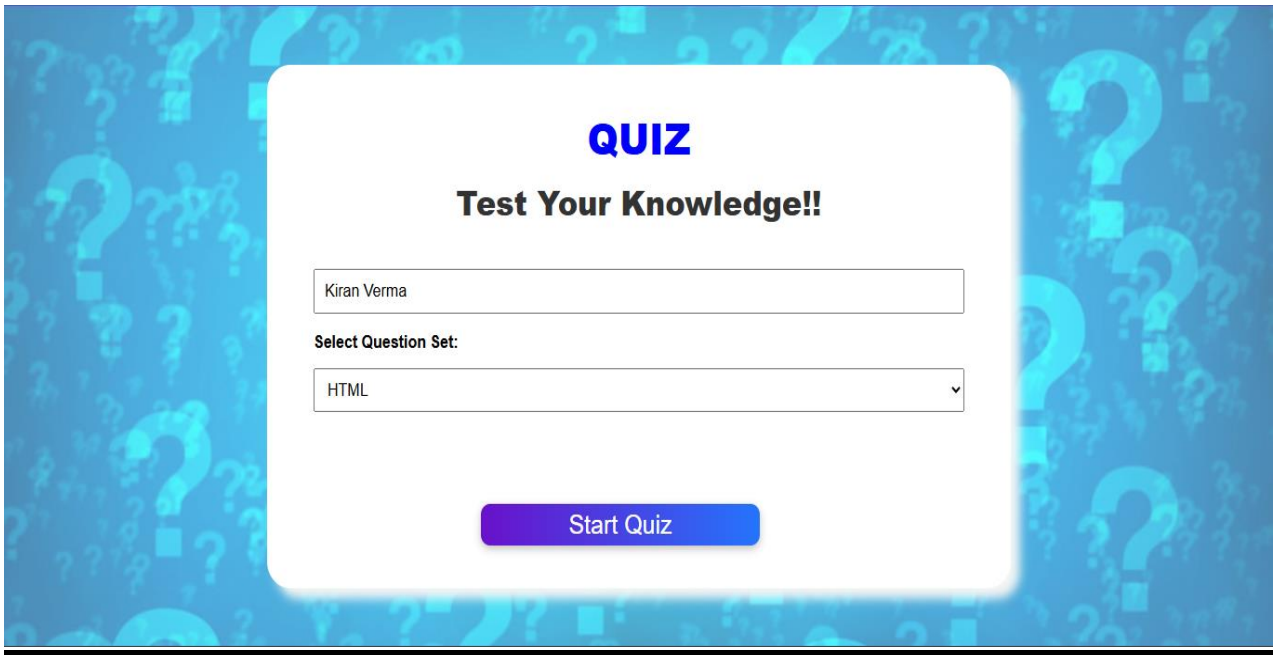
- Displays the final score
- Option to restart the quiz

Interface Design

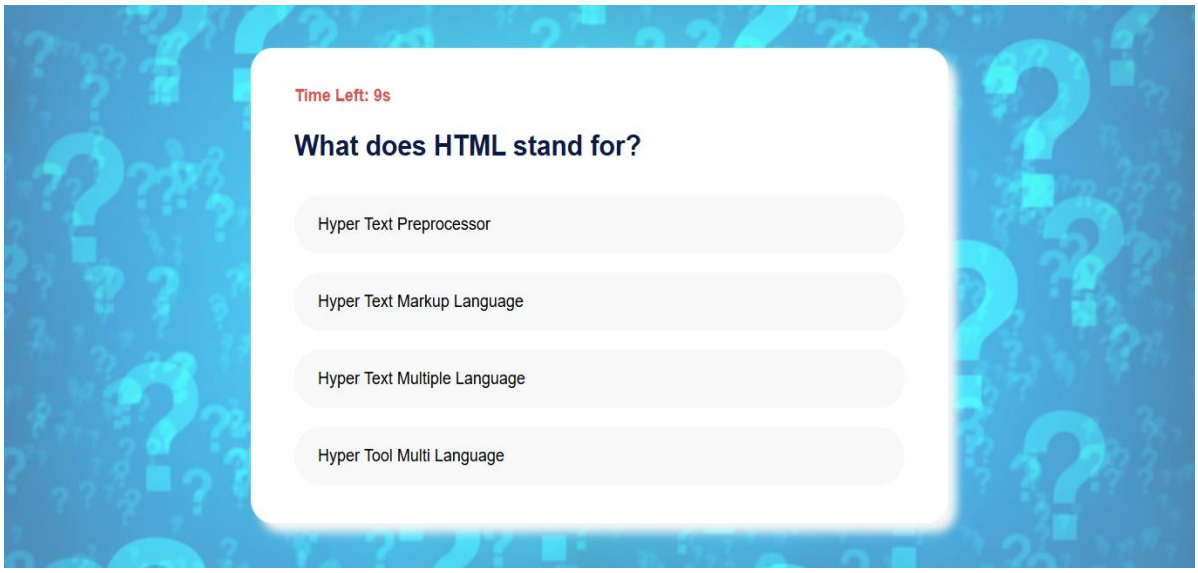
1.Home Page

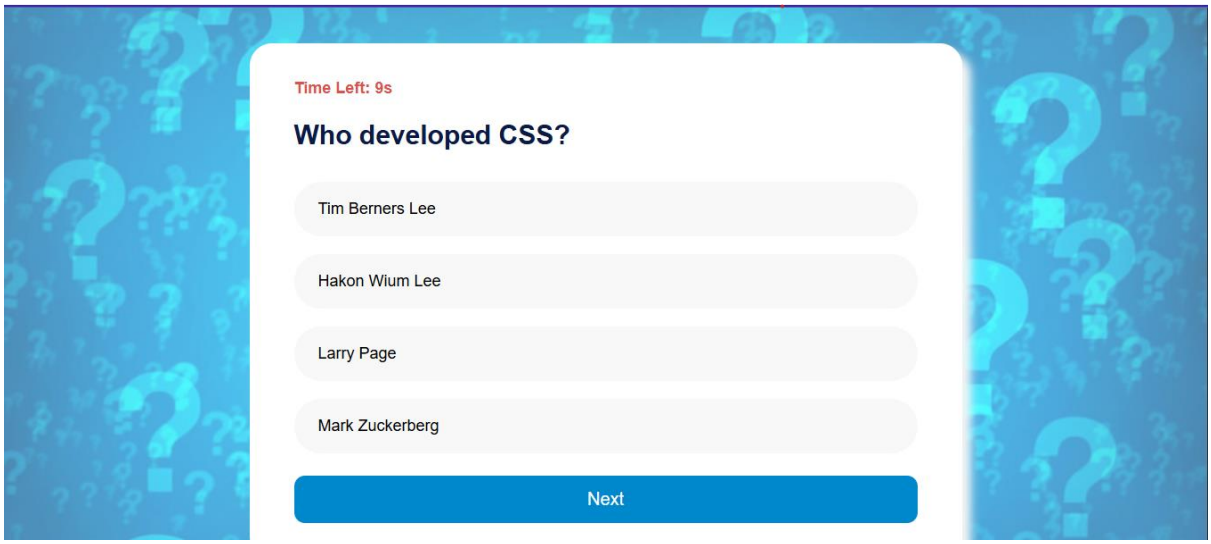


2.Login Page



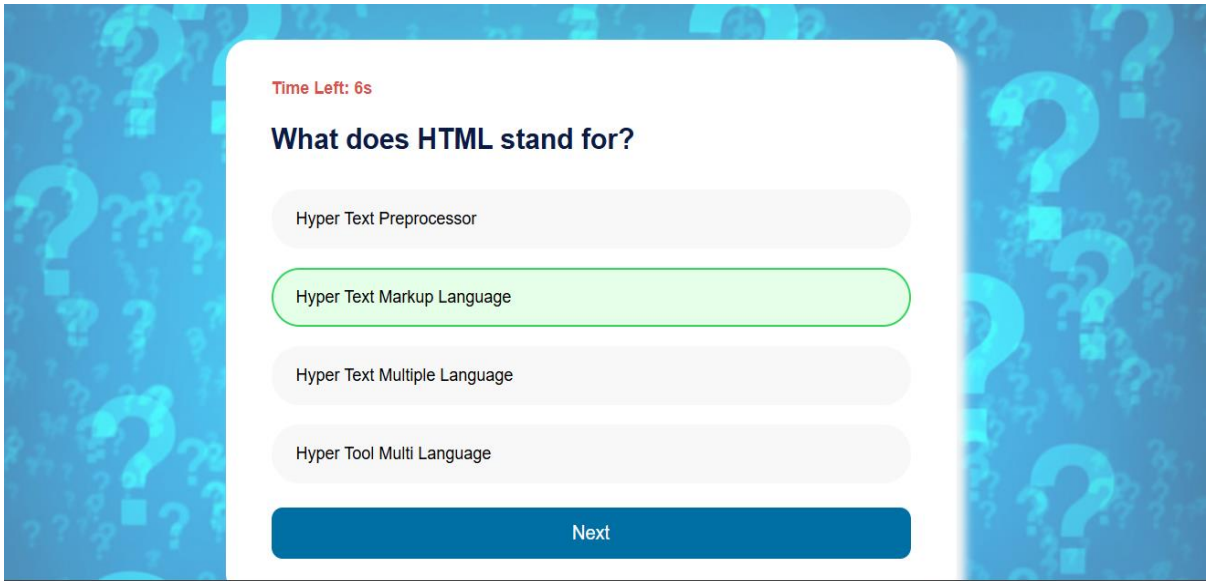
3.Question Page





Time Left: 10s
Who created JavaScript?
Brendan Eich
Tim Berners Lee
Larry Wall
Guido van Rossum
Next

4.Result:-


Time Left: 6s
What does HTML stand for?
Hyper Text Preprocessor
Hyper Text Markup Language
Hyper Text Multiple Language
Hyper Tool Multi Language
Next

Time Left: 3s

Who developed first version of HTML?

- Marc Andreessen
- Tim Berners-Lee
- Bill Gates
- Larry Page

Next

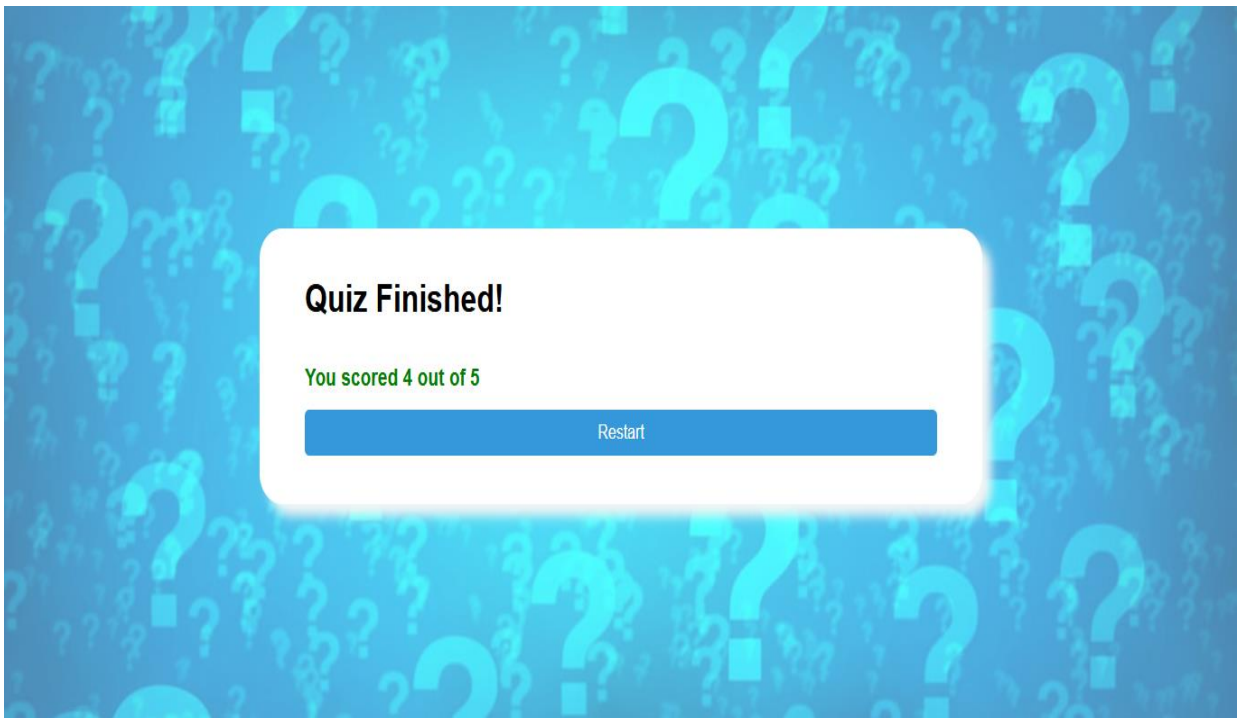
Time Left: 13s

Which is the correct way to write a comment in JavaScript?

- // This is a comment
- <!-- This is a comment -->
- /* This is a comment */
- Both A and C

Next

5.Final Score:-



CODE SNIPPETS

1.Question Data

```javascript
// 1. Quiz data
const quizSets = {
    html: [
        { q: "What does HTML stand for?", o: ["Hyper Text Preprocessor", "Hyper Text Markup Language", "Hyper Text Multiple Language", "Hyper Tool Multi Language"], a: 1 },
        { q: "Who developed first version of HTML?", o: ["Marc Andreessen", "Tim Berners-Lee", "Bill Gates", "Larry Page"], a: 1 },
        { q: "Which tag is used to define an unordered list?", o: ["<ol>", "<ul>", "<li>", "<dl>"], a: 1 },
        { q: "What is the correct HTML element for inserting a line break?", o: ["<br>", "<break>", "<lb>", "<newline>"], a: 0 },
        { q: "Which attribute is used for alternate text of an image?", o: ["alt", "title", "description", "src"], a: 0 }
    ],
    css: [
        { q: "What does CSS stand for?", o: ["Computer Style Sheets", "Creative Style Sheets", "Cascading Style Sheets", "Colorful Style Sheets"], a: 2 },
        { q: "Who developed CSS?", o: ["Tim Berners Lee", "Hakon Wium Lee", "Larry Page", "Mark Zuckerberg"], a: 1 },
        { q: "Which property changes background color?", o: ["background-color", "color", "bgcolor", "background"], a: 0 },
        { q: "Which CSS property controls text size?", o: ["font-size", "text-size", "font-style", "size"], a: 0 },
        { q: "How do you select an element with id 'header'?", o: ["#header", ".header", "header", "*header"], a: 0 }
    ],
    js: [
        { q: "What does JS stand for?", o: ["JavaScript", "Java Style", "Java Sheet", "Just Script"], a: 0 },
        { q: "Who created JavaScript?", o: ["Brendan Eich", "Tim Berners Lee", "Larry Wall", "Guido van Rossum"], a: 0 },
        { q: "Which is the correct way to write a comment in JavaScript?", o: ["// This is a comment", "<!-- This is a comment -->", "/* This is a comment */", "Both A and C"],
        { q: "Which symbol is used for strict equality?", o: ["==", "===", "=", "!="], a: 1 },
        { q: "How do you declare a variable in JavaScript?", o: ["var x = 5;", "variable x = 5;", "v x = 5;", "declare x = 5;"], a: 0 }
    ]
};
```

2.Load Question:-

```javascript
// 4. Load question
function loadQuestion() {
    const q = quiz[index];
    document.getElementById("question").innerText = q.q;

    const optionsDiv = document.getElementById("options");
    optionsDiv.innerHTML = "";

    q.o.forEach((opt, i) => {
        const div = document.createElement("div");
        div.classList.add("option");
        div.innerText = opt;
        div.onclick = () => checkAnswer(i);
        optionsDiv.appendChild(div);
        startTimer();

    });
```

3.Start Timer:-

```
    function startTimer() {
    clearInterval(timer);  // stop previous timer
    timeLeft = 15;
    document.getElementById("timer").innerText = `Time Left: ${timeLeft}s`;

    timer = setInterval(() => {
        timeLeft--;
        document.getElementById("timer").innerText = `Time Left: ${timeLeft}s`;

        if (timeLeft <= 0) {
            clearInterval(timer);
            autoSubmit();
        }
    }, 1000);
}
```

4.Check Answer:-

```
// 5. Check answer
function checkAnswer(selected) {
    clearInterval(timer);

    const optionsDiv = document.getElementById("options");
    const children = optionsDiv.children;

    for (let i = 0; i < children.length; i++) {
        children[i].onclick = null;
        if (i === quiz[index].a) children[i].classList.add("correct");
    }

    if (selected !== quiz[index].a) children[selected].classList.add("wrong");
    else score++;

    document.getElementById("nextBtn").style.display = "block";
}

```

TESTING

The system was tested using:

1. Unit Testing

- Timer function
- Answer checking

2. Functionality Testing

- Category selection
- Navigation between questions

3. UI Testing

- Button styling
- Responsive layout

All modules worked correctly.

CONCLUSION

The project achieves its objective of creating a simple, interactive, and user-friendly quiz system using HTML, CSS, and JavaScript. It is responsive, visually appealing, and efficiently handles quiz logic, making it suitable for educational and training purposes.

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